



Building an enterprise-ready generative AI platform on AWS

AWS Prescriptive Guidance



AWS Prescriptive Guidance: Building an enterprise-ready generative AI platform on AWS

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Table of Contents

Introduction	1
Intended audience	2
Objectives	2
Layered approach	4
Layer 1: Data and infrastructure	6
Foundation infrastructure for API-based implementations	6
Vector storage and retrieval infrastructure	7
High-performance compute infrastructure for model training and fine-tuning	7
Implementation recommendations	8
Layer 2: Foundation model access	9
Model experimentation and customization	9
Model evaluation	9
Objectives of custom model evaluation	10
Choosing model evaluation metrics	10
Model evaluation techniques in practice	12
Implementation recommendations	12
Layer 3: Security and governance	14
Core security disciplines	14
Recommended security controls	15
Security scoping matrix	16
Implementation recommendations	17
Layer 4: Repeatable application patterns	18
Generative AI-powered assistants	18
RAG chat-based assistants	19
Generative AI agents	20
Intelligent document processing	21
Content generation and summarization	22
Code generation and analysis	23
Multimodal AI applications	23
Implementation recommendations	24
Best practices	25
Organizational structure and governance	25
AI center of excellence	25
Model governance committee	25

Standardization and technical excellence	26
Pattern library and tooling	26
Enterprise-wide access framework	26
Process implementation	26
Service management	26
Model selection and evaluation	27
Performance monitoring and optimization	27
Security and compliance	27
Implementation recommendations	28
Next steps	29
Resources	31
AWS Prescriptive Guidance	31
Strategy resources	31
RAG and chat-based assistants	31
Coding assistants	31
Document processing	31
AWS service documentation	32
Other AWS resources	32
Other resources	32
Document history	33

Building an enterprise-ready generative AI platform on AWS

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It's common to hear that prototypes are easy, demos are cool, but production is hard. This is especially true with generative AI applications. This strategy document provides comprehensive guidance to help organizations implement secure, scalable, and effective generative AI platforms on AWS. It outlines a four-layer architectural approach that encompasses reliable infrastructure, foundation model selection, robust security and governance, and repeatable application patterns.

AI can help organization deliver differentiated products and services. Among the different domains of AI, generative AI has emerged as a powerful catalyst. It redefines how enterprises can simulate scenarios, optimize operations, and deliver compelling digital experiences to their customers.

While many organizations are creating numerous generative AI prototypes, they struggle to scale these into production-ready solutions. To facilitate widespread generative AI adoption, enterprises must address several critical challenges:

- Infrastructure readiness and scalability
- Security and compliance requirements
- Responsible AI implementation
- Integration with existing applications and processes
- Protection of sensitive data and intellectual property
- Cost optimization and return on investment (ROI) measurement

Additionally, you need to think about the ROI of solutions that are powered by generative AI. How can you integrate generative AI into existing applications? How can you make sure that AI-generated recommendations are safe and reliable? How can you help protect sensitive data and intellectual property when using existing foundation models or fine-tuned models?

By developing enterprise-ready generative AI capabilities, you can overcome these challenges. This strategy document offers a roadmap to help you understand and accelerate generative AI initiatives while maintaining compliance with security and ethical standards. It also describes how you can use AWS services, such as Amazon Bedrock and Amazon Q, to achieve your target business outcomes. These services can power everything from customer service automations to developer

productivity tools. They can also help you implement chat-based assistants and AI agents that transform your business operations.

Note

Before implementing this guidance, we recommend that you assess your organization's generative AI readiness by using the [Generative AI workload assessment](#). This strategy document builds upon the established AWS best practices and patterns for enterprise generative AI adoption that are documented in this guide.

Intended audience

This strategy document is intended for chief executive officers (CEOs), chief technology officer (CTOs), chief information officers (CIOs), and other senior technology leaders who are responsible for driving AI innovation and digital transformation. To use this strategy document, you should be familiar with business and technical management concepts, such as enterprise architectures, cloud computing models, IT governance frameworks, strategic technology planning, and digital transformation initiatives.

Objectives

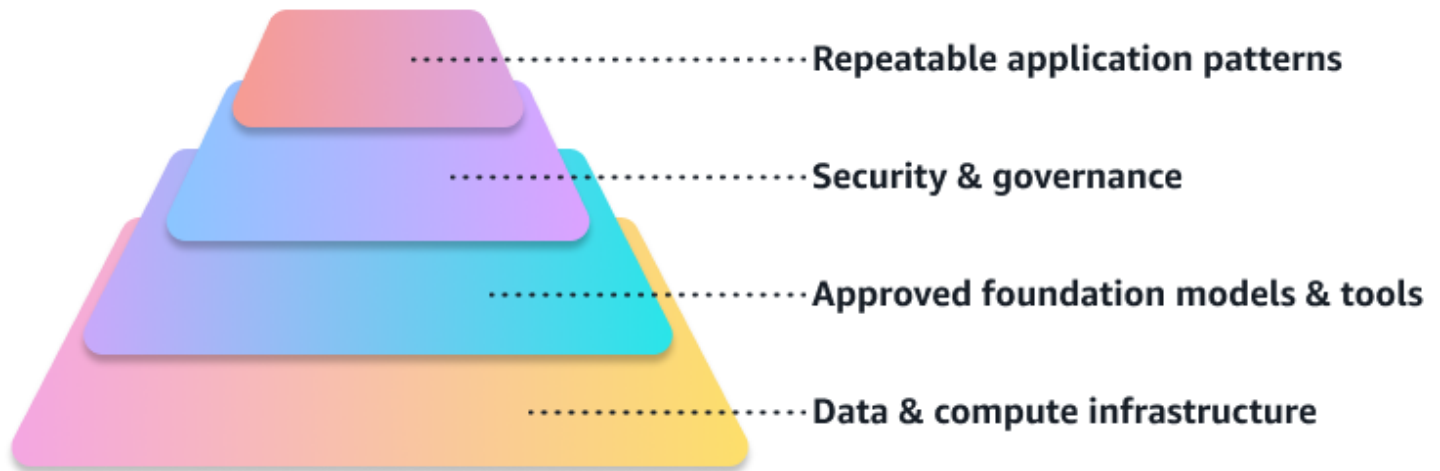
By implementing the recommendations in this strategy document, organizations can achieve the following targeted business outcomes:

- Transform prototypes into production-ready generative AI solutions that deliver measurable business value
- Help distributed teams experiment faster while maintaining security and governance
- Establish enterprise-wide access to approved foundation models with proper controls and governance
- Reduce development time and costs through reusable application patterns and standardized tooling
- Accelerate innovation through secure, self-service access to generative AI capabilities
- Implement responsible AI practices that promote ethical use and compliance
- Create a scalable framework for evaluating and adopting new foundation models
- Build trust in AI-generated content through proper validation and control mechanisms

- Optimize operational costs while maintaining high performance and security standards
- Seamlessly integrate generative AI capabilities into existing enterprise applications

Adopting a layered approach for generative AI development

To build a robust and scalable generative AI platform, organizations need a comprehensive environment that enables innovation while maintaining control and security. This strategy document recommends an enterprise-ready environment that consists of four essential layers, which are shown in the following image.



Each of the following layers plays a crucial role in successful generative AI adoption and scaling across the enterprise:

1. **Data and compute infrastructure** – Reliable infrastructure provides the foundation for developing, training, and deploying generative AI applications. This layer makes sure that organizations have the necessary computational resources and data management capabilities to support various use cases, from small experiments to large-scale production deployments. For Amazon Elastic Compute Cloud (Amazon EC2), consider using [AWS Trainium](#), [AWS Inferentia](#), [UltraClusters](#), [Elastic Fabric Adapter](#), [Capacity Blocks for ML](#), [AWS Nitro System](#), and [AWS Neuron](#).
2. **Approved foundation models and tools** – This layer allows your organization to access approved, pretrained models while maintaining security and compliance. Teams can experiment with different models, evaluate their effectiveness, and select the most appropriate ones for specific use cases. You can use [Amazon Bedrock](#) to establish guardrails, build agents, customize models, or import your own custom model.
3. **Security and governance** – This layer helps you make sure that generative AI applications adhere to organizational policies, compliance requirements, legal and privacy requirements,

and ethical guidelines. In this layer, you implement consistent security controls and governance mechanisms across all generative AI initiatives.

4. **Repeatable application patterns** – This layer accelerates development and promotes consistency by providing standardized templates and best practices for common generative AI use cases. Providing repeatable patterns helps prevent duplication of effort and enables rapid scaling of successful solutions. Examples of repeatable patterns include intelligent document processing, Retrieval Augmented Generation (RAG), chat-based assistants, and agentic workflows.

This layered approach provides the following key benefits:

- It clearly separates the different aspects of generative AI implementation.
- It promotes consistent security and governance across all generative AI initiatives.
- It helps you build scalable infrastructure that grows with organizational needs.
- It standardizes approaches that accelerate development.
- It reduces risk through proven patterns and practices.

By implementing these four layers, organizations can create a solid foundation for their generative AI initiatives while maintaining the flexibility to adapt to new requirements and opportunities.

Layer 1: Building reliable data and compute infrastructure for generative AI

For developing generative AI applications, particularly if training or fine-tuning a foundation model is necessary, a robust data foundation and compute infrastructure is critical. As enterprises embark on their generative AI journey, they need infrastructure that can support the entire machine learning lifecycle. They also need to strike the right balance between performance, cost, and operational efficiency.

Reliable generative AI infrastructure should offer the following three key components: the foundational infrastructure, the vector storage and retrieval infrastructure, and the compute infrastructure. Together, these components provide the flexibility to serve the needs of any project, regardless of its scale, requirements, or environment.

This section contains the following topics:

- [Foundation infrastructure for API-based implementations](#)
- [Vector storage and retrieval infrastructure](#)
- [High-performance compute infrastructure for model training and fine-tuning](#)
- [Implementation recommendations](#)

Foundation infrastructure for API-based implementations

Most organizations begin their generative AI journey by using pretrained foundation models through APIs. [Amazon Bedrock](#) provides serverless access to leading foundation models through a unified API. This helps you experiment and deploy generative AI applications without managing complex infrastructure. For these implementations, you need the following:

- [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) general-purpose instances for running your applications.
- [Amazon Simple Storage Service \(Amazon S3\)](#) for storing application data and outputs.
- [Amazon CloudWatch](#) for monitoring and logging your application's performance and usage.
- (Optional) [AWS Lambda](#) for serverless compute when building event-driven generative AI applications.

Vector storage and retrieval infrastructure

As your generative AI applications mature, you'll likely need to enhance them with domain-specific knowledge and context. With Retrieval Augmented Generation (RAG), the foundation model references an authoritative data source that is outside of its training data sources, such as your organization's data or documents, before generating a response. For more information, see [Retrieval Augmented Generation options and architectures on AWS](#).

[Amazon Bedrock Knowledge Bases](#) provides a fully managed solution for building RAG applications. It helps you to securely connect your enterprise data with foundation models. For additional flexibility, you can use [Amazon OpenSearch Service](#) with vector search capabilities, or you can use [Amazon Relational Database Service \(Amazon RDS\) for PostgreSQL](#) with the [pgvector](#) extension to store vector embeddings.

For efficient vector operations, consider using Amazon EC2 compute-optimized instances for embedding generation. Also consider caching frequently accessed embeddings by using [Amazon ElastiCache](#) to optimize performance and reduce costs.

For more information about vector databases and caching, see [What is a vector database?](#) and [Choosing an AWS vector database for RAG use cases](#).

High-performance compute infrastructure for model training and fine-tuning

For organizations that are ready to customize foundation models, AWS offers comprehensive infrastructure for model training and fine-tuning. You can use Amazon Simple Storage Service (Amazon S3) as low cost, scalable, and highly durable object storage for building data platforms and for storing training data and trained models. In addition, [AWS Glue](#) is a serverless data integration service that can help you prepare data for model training.

For the training infrastructure, [Amazon SageMaker AI](#) offers a fully managed machine-learning environment with the tools and workflows that you need to build, train, and deploy models. Using SageMaker AI can significantly reduce your operational overhead. Consider using [accelerated computing instances](#), such as the G and P instance families. These instance families provide access to the latest GPUs from NVIDIA for machine learning training and inference. You can also use [AWS Trainium](#), a purpose-built machine learning accelerator that speeds up training times by up to 50%, while lowering costs.

For truly large projects, you can use [Amazon EC2 UltraClusters](#). UltraClusters consist of thousands of accelerated Amazon EC2 instances that are co-located in a given AWS Availability Zone and interconnected. UltraClusters can scale to thousands of GPUs or machine learning accelerators. They deliver exa-floating point operations per second (exaflops) of aggregate compute capacity, which reduces training times and can reduce time-to-solution from weeks to a few days.

Implementation recommendations

Consider the following recommendations for setting up a scalable and cost-effective infrastructure for your generative AI projects:

- For quick experimentation and initial deployments, start with Amazon Bedrock and use general-purpose compute instances for your applications.
- As your needs evolve, implement vector storage solutions by using Amazon Bedrock Knowledge Bases or Amazon OpenSearch Service. Scale your infrastructure accordingly.
- For advanced customization, standardize and automate the provisioning of secure and governed machine learning environments to support the requirements of distributed teams. For more information, see [Setting up secure, well-governed machine learning environments on AWS](#) (AWS blog post).
- Adopt machine learning operations (MLOps) to automate and standardize processes across the machine learning lifecycle. These processes include model development, testing, integration, release, and infrastructure management. For more information, see [What is MLOps?](#)
- For small-scale experiments or proof of concepts, start with Amazon SageMaker AI and general-purpose compute instances. As you scale to large production deployments, consider Amazon EC2 accelerated computing instances for maximum performance.
- Use [managed spot training](#) in SageMaker AI to optimize the cost of training models by up to 90% compared to on-demand instances. SageMaker AI manages the spot interruptions on your behalf.

Layer 2: Approved set of foundation models and tools

As organizations navigate the early stages of generative AI adoption, they quickly realize that no single model can address all use cases effectively. Different models excel in various domains and tasks, and enterprises need to balance capability, cost, and performance for each specific application. This reality drives the need for a flexible, yet controlled, approach to foundation model access.

This section contains the following topics:

- [Model experimentation and customization](#)
- [Model evaluation](#)
- [Implementation recommendations](#)

Model experimentation and customization

[Amazon Bedrock](#) is designed to help you experiment with various foundation models, and it supports scalable production deployments. With Amazon Bedrock [Knowledge Bases](#), you have a fully managed solution to build end-to-end Retrieval Augmented Generation (RAG) workflows. Amazon Bedrock also supports managed agents that can run complex tasks without code, from booking travel to managing inventory. Because it's serverless, Amazon Bedrock reduces infrastructure management concerns and integrates securely with other AWS services.

Additionally, you can use Amazon Bedrock to [privately customize the foundation models](#) with your own proprietary data and make them securely available to the users within your organization. For more information, see [Customize your model to improve its performance for your use case](#) in the Amazon Bedrock documentation. For access management, AWS Identity and Access Management (IAM) integrates with Amazon Bedrock so that you can configure fine-grained access control. The controls determine which users can enable and access specific models. For more information, see [Layer 3: Security and governance](#) in this guide.

Model evaluation

As your organization progresses from generative AI prototypes to production, it's essential to establish rigorous evaluation processes for foundation models. Although open benchmarks offer

general insights into model performance, they often fall short when determining a model's suitability for specific enterprise needs. Tailored evaluation strategies help users select the most appropriate model that aligns with the unique requirements of your organization.

Objectives of custom model evaluation

A custom evaluation approach helps organizations do the following:

- **Assess performance on business-relevant tasks** – Evaluate how well models handle tasks directly related to your enterprise use cases.
- **Identify potential biases and limitations** – Detect areas where models might exhibit biases or fail so that you can make sure the model is suitable for real-world deployment.
- **Compare models with relevant metrics** – Compare different models by using metrics that align you're your organizational priorities and objectives.
- **Make informed model selection** – Make data-driven decisions about model selection, fine-tuning, and deployment to production environments.

Choosing model evaluation metrics

Effective model evaluation draws on a mix of techniques that provide a holistic view of model performance. With these techniques, organizations can assess not only the technical accuracy of a model but also its alignment with enterprise-specific needs. To evaluate a model, you combine quantitative and qualitative metrics to determine performance, identify biases, and choose a model. Organizations should use both ground truth-based metrics (with reference data) and flexible metrics (without reference data) to gain a comprehensive view of model suitability. In AI, *ground truth* refers to data that is factual and is withheld during model training so that you can use it for model evaluation.

Use ground truth-based metrics if reference data exists. These metrics provide concrete, quantitative assessments. Meanwhile, techniques without ground truth can offer flexibility. These techniques help you evaluate models on dimensions such as readability and completeness, even when predefined correct answers aren't available.

Metrics based on ground truth data

If reference data is available, ground truth-based metrics can provide quantitative assessments. The following metrics provide quantitative performance assessments:

- **ROUGE-L score** – Recall-oriented understudy for gisting evaluation (ROUGE) for longest common subsequence (LCS), also known as *ROUGE-L*, measures the longest common subsequence between generated and reference texts. The metric assesses the coherence and content overlap.
- **Cosine similarity** – This metric evaluates semantic similarity between texts. It provides insights into the model's contextual understanding.
- **METEOR score** – Metric for evaluation of translation with explicit ordering (METEOR) scoring combines word alignment and semantic matching to provide a balanced assessment of content accuracy and meaning.
- **Binary similarity** – This metric checks for exact matches, making it particularly useful for tasks that require precise outputs, such as command generation.
- **LLM-as-a-judge score** – This metric uses another large language model (LLM) to rate similarity on a defined scale. It offers a nuanced evaluation of quality.

Amazon SageMaker Clarify and Amazon Bedrock include features to help you evaluate models. They can automate model evaluation jobs so that you can quantify model risks and response quality. For more information, see [Evaluate, explain, and detect bias in models](#) and [Evaluate the performance of Amazon Bedrock resources](#).

Metrics without ground truth data

When reference data is unavailable, or as a complementary approach, you can use the *LLM-as-a-judge technique*. It uses a separate model to evaluate the outputs of a generative AI solution based on dimensions that are important to the business. This technique provides flexibility for assessing various model qualities. For more information, see [Evaluate model performance using another LLM as a judge](#) in the Amazon Bedrock documentation.

You can use computed metrics to evaluate how effectively a Retrieval Augmented Generation (RAG) system retrieves relevant information from your data sources, and how effective the generated responses are in answering questions. The results of a RAG evaluation allow you to compare different Amazon Bedrock Knowledge Bases and other RAG sources, and then to choose the best Knowledge Base or RAG system for your application. For more information, see [Evaluate the performance of RAG sources](#) in the Amazon Bedrock documentation.

Model evaluation techniques in practice

To effectively match enterprise needs with model capabilities, use a robust set of evaluation criteria to direct the model evaluation process. These criteria cover a spectrum of priorities, from correctness and completeness to factuality and sensitive data handling. Each criterion aligns with specific techniques to deliver actionable insights. For instance, when factual accuracy is critical, metrics like ROUGE-L score or cosine similarity provide concrete, quantifiable benchmarks. Conversely, use techniques such as LLM-as-a-judge to evaluate readability and freshness. These techniques offer flexible, qualitative insights tailored to organizational standards. Key dimensions for evaluating models include:

- **Correctness** – Validates the accuracy of information provided
- **Completeness** – Verifies the depth and coverage of responses
- **Readability** – Assesses clarity and ease of understanding
- **Freshness of information** – Checks that content is relevant and current
- **Sensitive data suppression** – Checks for proper handling of confidential information
- **Accuracy** – Measures alignment with factual information
- **Coherence** – Examines logical flow and consistency
- **Factuality** – Verifies the truthfulness of the content
- **Comprehensiveness** – Assesses the scope and thoroughness of coverage

By anchoring model evaluation techniques in these clear criteria, you can thoroughly vet models for their specific purposes. This structured approach aligns models with enterprise goals, compliance requirements, and user expectations. It also lays a strong foundation for deploying generative AI applications at scale in production environments.

Implementation recommendations

To establish an effective foundation model strategy, consider the following recommendations:

- Form a model governance committee that has clear roles, responsibilities, and decision-making processes.
- Develop evaluation criteria and scoring mechanisms for the assessment of foundation models before they are available for use across the organization.

- Remember that the largest foundation model is not necessarily always the best model for your use case. Begin proof-of-concept development with top-tier models to validate business value, and then systematically evaluate smaller models for cost optimization.
- Develop dashboards to track key metrics, such as inference latency, throughput, error rates, and cost per inference.
- Provide clear guidance to teams about how to select the right model for their use case, including experimentation processes and evaluation criteria.

Layer 3: Security and governance for generative AI platforms on AWS

A robust security and governance framework is essential for scaling generative AI adoption across the enterprise. We recommend a platform-centric approach. This approach makes sure that all generative AI powered applications, irrespective of which team builds them, benefit from a default set of security and responsible AI guardrails.

This section contains the following topics:

- [Core security disciplines](#)
- [Recommended security controls](#)
- [Security scoping matrix](#)
- [Implementation recommendations](#)

Core security disciplines

[OWASP top 10 for LLM applications](#) describes additional security considerations for generative AI workloads. However, many traditional security disciplines remain crucial. Security considerations for generative AI workloads include the following:

- **Governance and compliance** – Create the necessary policies, procedures, and reporting to empower the business while minimizing risk. For generative AI applications, this encompasses guidelines for model selection, data usage, and output validation.
- **Legal and privacy** – Meet the specific regulatory, legal, and privacy requirements for using or creating generative AI solutions. Organizations must carefully consider data protection laws, intellectual property rights, and industry-specific regulations when implementing generative AI solutions.
- **Risk management** – Identify potential threats to generative AI solutions and recommended mitigations. This includes addressing risks such as data poisoning, prompt injection attacks, or unintended biases in model outputs.
- **Controls** – Implement security controls that mitigate risk. For generative AI applications, this includes input sanitization, output filtering, and access controls for model usage. For more information, see Recommended security controls in this guide.

- **Resilience** – Architect generative AI solutions to maintain availability and meet business service-level agreements (SLAs). This includes considerations for model redundancy, fallback mechanisms, and scalability under varying load conditions.

Recommended security controls

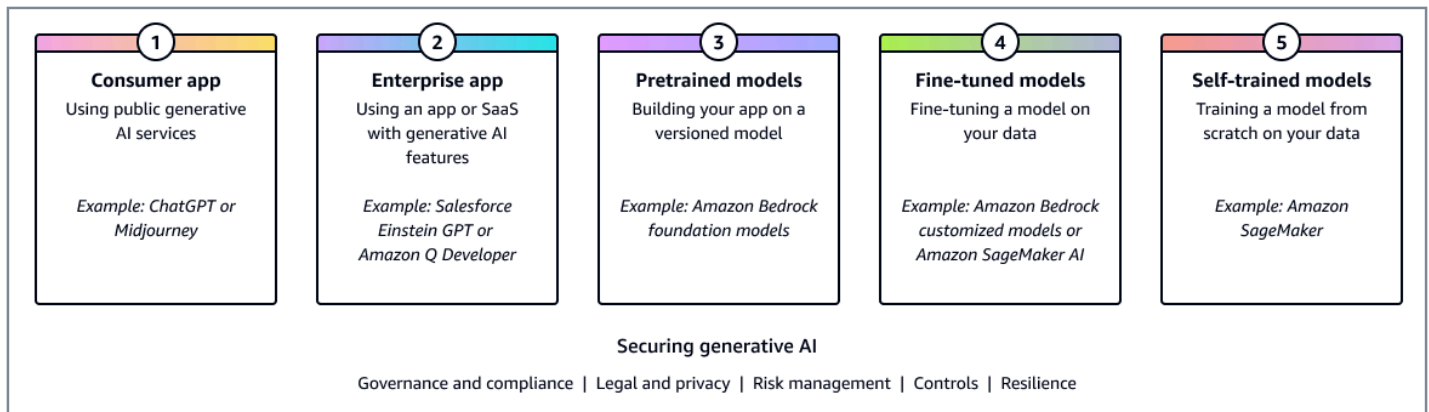
Defense-in-depth is a cybersecurity approach that uses layers of security controls to protect a system, network, or workload. Different layers help detect and stop attacks if one layer fails. At the minimum, we recommend the following controls to adopt a defense-in-depth approach for generative AI workloads and their environments:

- **Identity and access management** – AWS services, including Amazon Bedrock and Amazon SageMaker AI, natively integrate with AWS Identity and Access Management (IAM). IAM provides granular control over who can perform actions on your AWS accounts and resources, such as subscribing to foundation models or running inference on a model. We recommend that you create multiple IAM [roles](#) with granular permissions. For example, you might create the following roles:
 - *Evaluation role* – Users with this role can evaluate foundation models in a sandbox environment as new models are added to Amazon Bedrock. These users should coordinate with your legal and procurement teams before activating subscriptions that allow broader access.
 - *General access role* – Users with this role can access foundational models for standard usage.
 - *Fine-tuned model access role* – Users with this role can access models that are fine-tuned with your proprietary data.
 - *Specialized model access role* – Users with this role can access to cutting-edge, high-cost models for specific use cases.
- **Private network access through AWS PrivateLink** – Use [AWS PrivateLink](#) with Amazon Bedrock and Amazon SageMaker AI to invoke models from within your VPC and on-premises network. This helps keep sensitive data within your private network.
- **Guardrails for Amazon Bedrock** – Implement [guardrails](#) to manage and filter the requests and responses to and from foundation models. This adds an additional layer of control.
- **Securely store invocation logs** – To make sure that use meets your compliance needs, you can store all requests and responses to an Amazon S3 bucket or to Amazon CloudWatch Logs. For more information, see [Monitor model invocation using CloudWatch Logs and Amazon S3](#) in the Amazon Bedrock documentation.

- **Audit and track access to foundation models** – Use Amazon CloudWatch, a service that records actions taken by a user, role, or an AWS service. CloudTrail captures all API calls for Amazon Bedrock as events. For more information, see [Monitor Amazon Bedrock API calls using CloudTrail](#) and [Logging Amazon SageMaker AI API calls using AWS CloudTrail](#).
- **Follow OWASP top 10 for LLM security** – Adhere to OWASP's top 10 security risks for LLM applications, and make sure that you have clearly identified strategies and security controls to architect layered defenses. For more information, see [Architect defense-in-depth security for generative AI applications using the OWASP Top 10 for LLMs](#) (AWS blog post).

Security scoping matrix

The specific security controls that you need depends on the nature of your generative AI application. The [AWS Generative AI Security Scoping Matrix](#) (AWS blog post) provides a framework for understanding the security requirements based on your deployment scope. The following image shows the generative AI security scoping matrix, which is a mental model to classify use cases.



The matrix helps organizations identify the appropriate security controls based on factors such as:

- Whether the application is consumer-facing or internal to the enterprise
- The use of pretrained models compared to fine-tuning
- The sensitivity of the data being processed
- The criticality of the application to business operations

Implementation recommendations

To implement a comprehensive security and governance framework for your generative AI workloads, consider the following recommendations:

- Establish clear policies and procedures for model access and usage.
- Implement security controls at multiple layers, such as at the network, application, and data layers.
- Perform regular security assessments and compliance audits.
- Maintain detailed documentation for security measures and controls.
- Provide security awareness training for teams that work with generative AI.
- Regularly review and update security controls as threats evolve.

Layer 4: Repeatable application patterns for common generative AI use cases

As enterprises adopt generative AI, clear patterns have emerged for how organizations are using this technology to drive business value. These common use cases demonstrate proven approaches to applying generative AI across different business functions. These use cases can help organizations identify and prioritize their own implementation opportunities.

By understanding these established use cases, organizations can better evaluate where generative AI can deliver the most immediate impact while building a foundation for broader adoption. This section describes key areas where enterprises are successfully deploying generative AI solutions.

This section contains the following topics:

- [Generative AI-powered assistants](#)
- [RAG chat-based assistants](#)
- [Generative AI agents](#)
- [Intelligent document processing](#)
- [Content generation and summarization](#)
- [Code generation and analysis](#)
- [Multimodal AI applications](#)
- [Implementation recommendations](#)

Generative AI-powered assistants

Generative AI-powered assistants offer personalized, context-aware support across various business functions. These tools are ideal for power users. They supercharge the output of those employees who learn to integrate them into their workflow. By using language models, these assistants enhance efficiency, productivity, and user satisfaction across organizations.

AWS offers [Amazon Q Business](#), an enterprise-grade AI assistant that's ready to use and highly customizable. Implementation involves tailoring its knowledge base with organization-specific data to drive accurate, relevant responses. Integration with customer relationship management (CRM) and enterprise resource planning (ERP) systems enhances its utility and promotes smooth data flow and full functionality across the enterprise.

For software development teams, AWS provides [Amazon Q Developer](#). This coding assistant is designed to boost productivity by assisting with code generation, bug detection, refactoring suggestions, and documentation creation. When integrated seamlessly into integrated development environments (IDEs), developers can access its features directly within their usual coding environment, which streamlines workflows and reduces context switching. For more information, see [Code generation and analysis](#) in this strategy document.

Common use cases for generative AI-powered assistants include the following:

- Executive assistants that support schedule management, report summarization, and brief preparation.
- HR assistants that support employee onboarding, answer policy-related questions, and help with performance management.
- Sales enablement tools that deliver real-time product information and competitive insights.

To maximize the effectiveness of generative AI-powered assistants, consider starting with a targeted deployment, such as within a specific department. This helps development teams to evaluate the impact and gather user feedback before rolling it out more broadly across the organization. This phased approach smooths the adoption process and optimizes the assistant's alignment with enterprise needs.

For more information about building enterprise generative AI-powered assistants, see [Develop a fully automated chat-based assistant using Amazon Bedrock agents and knowledge bases](#).

RAG chat-based assistants

Out-of-the-box foundation models, while powerful, are constrained by their pretraining data. They lack knowledge of recent events beyond their training cutoff date and have no understanding of company-specific processes or custom data. This limitation makes them unsuitable for many business use cases that require knowledge of the organization.

Customizing models through fine-tuning is a complex and expensive process, and collecting high-quality training data for it can be a challenge. Retrieval Augmented Generation (RAG) addresses these limitations by combining large language models (LLMs) with an organization's proprietary data. RAG approaches create AI-powered interactions that are informed by specific domain knowledge. For more information about RAG and fine-tuning, see [Generative AI options for querying custom documents](#).

Common use cases for RAG chat-based assistants include the following:

- Customer support systems that have access to product documentation and FAQs
- Internal knowledge management tools that quickly retrieve company-specific information
- Legal or compliance chat-based assistants that reference organizational policies and regulations

AWS simplifies the implementation of RAG chat-based assistants through [Amazon Bedrock Knowledge Bases](#). This service helps you set up a vector database to query and retrieve data from custom documents. Key features include integration with Amazon Simple Storage Service (Amazon S3) for data storage, compatibility with third-party services (such as Microsoft SharePoint and Atlassian Confluence), efficient vector search capabilities for relevant document retrieval, and integration with foundation models (such as Anthropic Claude) for building conversational chat-based assistants. To begin, identify a well-defined, limited-scope knowledge base and use it to create a proof-of-concept RAG chat-based assistant. As implementations scale, pay attention to data freshness and the computational costs of large-scale retrieval operations.

For more information about building RAG chat-based assistants, see the following:

- [Retrieval Augmented Generation options and architectures on AWS](#)
- [Deploy a RAG use case on AWS by using Terraform and Amazon Bedrock](#)
- [Develop advanced generative AI chat-based assistants by using RAG and ReAct prompting](#)

Generative AI agents

Generative AI agents represent a significant advancement in automating complex business processes. These agents can handle sophisticated workflows that traditionally required human intervention by breaking down tasks and interacting with various systems.

Typical applications of generative AI agents include automated workflow orchestration for multi-step business processes, intelligent virtual assistants for cross-departmental task completion, and data analysis and report generation agents that can query multiple databases.

AWS provides tools, such as [Amazon Bedrock Agents](#), to help you create and deploy AI agents. Implementation involves defining custom actions and API integrations that are specific to your business needs. You also need to set up robust error handling and logging mechanisms. You can use [AWS Lambda](#) for serverless function execution, [Amazon API Gateway](#) to manage APIs, and [AWS Step Functions](#) for workflow orchestration.

Organizations can start by identifying a repetitive, multi-step process and mapping out its decision points and system interactions. This serves as a blueprint for the initial generative AI agent. Key considerations include validating proper error handling and maintaining transparency in decision-making processes.

Intelligent document processing

Intelligent document processing (IDP) uses generative AI and natural language processing (NLP) to automatically extract, classify, and process information from various types of documents. IDP can transform unstructured document processing into automated, scalable workflows.

Modern IDP solutions can take two approaches: traditional, pipeline-based processing or direct, multimodal processing through foundation models. To build a traditional pipeline on AWS, you can combine [Amazon Textract](#) for optical character recognition (OCR) and document understanding with [Amazon Bedrock foundation models](#) for advanced analysis. Alternatively, for direct, multimodal processing through foundation models, you can use state-of-the-art foundation models (such as Anthropic Claude in Amazon Bedrock) to directly process images, PDFs, and scanned documents in a single step.

Organizations can choose their approach based on their specific needs. Traditional pipelines excel in high-volume processing and scenarios that require fine-grained control. Direct, multimodal processing through foundation models is ideal for complex documents that require improved accuracy through sophisticated contextual understanding.

Common use cases for IDP include the following:

- Claims processing that automates insurance claim document review, validation, and data extraction.
- Invoice processing that extracts line items, totals, and vendor information for accounts payable automation.
- Contract analysis that reviews legal documents in order to extract key terms, obligations, and expiration dates.
- Medical record processing that analyzes patient records to extract relevant medical information and coding.

Implementation typically uses event-driven architectures that use Amazon S3, AWS Lambda, and Amazon Simple Queue Service (Amazon SQS). When you upload a document to an Amazon

S3 bucket, it initiates processing workflows. Results are stored in databases, such as Amazon DynamoDB, or ingested into business systems through APIs. For optimal results, organizations should implement batching strategies, robust error handling, and audit trails. Start with a specific document type before expanding the scope.

This automation significantly reduces manual intervention, processing time, and human error. It also helps staff to focus on activities that require judgment and expertise.

For more information about document processing workflows, see [Automatically extract content from PDF files using Amazon Textract](#).

Content generation and summarization

Generative AI models excel at generating and summarizing content. This makes them valuable tools for various business applications. You can use the natural language processing capabilities of large language models to create or condense text-based information.

Common use cases for content generation and summarization include the following:

- Automated report generation from structured data
- Content creation for marketing materials, product descriptions, and social media posts
- Document summarization for research papers, legal documents, or lengthy reports

Implementation on AWS typically involves using Amazon Bedrock for text generation and summarization tasks. Amazon Comprehend can be employed for entity recognition and sentiment analysis, enhancing the context and quality of generated content.

Best practices for content generation and summarization applications include implementing human-in-the-loop processes for content review and quality control, using fine-tuned models for domain-specific content generation, and implementing content filters to ensure brand consistency and adherence to guidelines.

For example, you can use generative AI to analyze patient clinical data and generate summaries and clinical notes. For more information, see [Building a medical intelligence application with augmented patient data](#).

Code generation and analysis

Generative AI models have remarkable capabilities to understand and generate code across various programming languages. Your developers can use these capabilities to assist write, refactor, and analyze code.

Common use cases include the following:

- Automated code generation for boilerplate tasks
- Code refactoring and optimization suggestions
- Bug detection and security vulnerability analysis

AWS offers several services to help support this pattern. [Amazon Q Developer](#) provides AI-assisted code generation, helping developers write code faster and with fewer errors. [Amazon CodeGuru](#) offers intelligent code reviews and performance recommendations. You can integrate these services with AWS Lambda for serverless code execution and testing.

Best practices for code generation and analysis applications include integrating AI-assisted coding tools into existing development environments. Consider implementing strict code review processes to make sure that generated code meets quality standards. Use AI for initial code drafts but rely on human expertise for final implementation decisions.

For more information about building coding assistants, see [Use Amazon Q Developer as a coding assistant to increase your productivity](#).

Multimodal AI applications

Multimodal AI applications combine different types of data inputs and outputs, such as text, images, audio, and video. This pattern uses the ability of advanced generative AI models to understand and generate content across multiple modalities.

Common use cases for multimodal AI applications include the following:

- Visual question-answering systems for customer support or internal documentation
- Intelligent image and video content moderation
- Multimodal data analysis for business intelligence

On AWS, you can use Amazon Bedrock for multimodal foundation models. Additional services include [Amazon Rekognition](#) for image and video analysis, [Amazon Transcribe](#) for speech-to-text conversion, and [Amazon Polly](#) for text-to-speech conversion. These services can enhance the multimodal capabilities of the application.

Best practices for multimodal AI applications include ensuring high-quality, diverse datasets for training and fine-tuning multimodal models. Also, implement robust error handling for different input modalities, and carefully consider the ethical implications and potential biases in multimodal AI systems.

For more information about building multimodal applications, see [Document institutional knowledge from voice inputs by using Amazon Bedrock and Amazon Transcribe](#).

Implementation recommendations

To effectively implement repeatable application patterns for generative AI across your organization, consider the following recommendations:

- Begin with a pattern catalog that documents successful implementations and provides templates, reference architectures, and code samples that accelerate development.
- Establish a centralized pattern governance process to evaluate, approve, and maintain standard patterns in order to promote consistency and quality across the enterprise.
- Prioritize use cases based on business impact, technical feasibility, and alignment with strategic objectives.
- Create a cross-functional pattern development team that includes AI specialists, domain experts, and end users who can make sure that patterns address real business needs.
- Implement a "build once, reuse many times" approach by creating modular, composable components that can be assembled into different application patterns.
- Develop clear evaluation criteria for each pattern type in order to measure effectiveness. Include metrics for business value, user satisfaction, and technical performance.
- Establish a pattern maturity model that classifies patterns as experimental, proven, or enterprise-standard to guide teams in their selection process.
- Create comprehensive documentation that promotes successful adoption for each pattern, including implementation guides, limitations, and best practices.
- Conduct regular pattern reviews to identify opportunities for improvement, consolidation, or retirement as technology and business needs evolve.

Best practices for enterprise generative AI adoption and scaling

Successfully adopting and scaling generative AI across an enterprise requires a strategic balance of organizational structure, standardized processes, and technical capabilities. The following best practices draw from successful implementations across various organizations, providing a framework for effective enterprise-wide adoption.

This section contains the following topics:

- [Organizational structure and governance](#)
- [Standardization and technical excellence](#)
- [Process implementation](#)
- [Implementation recommendations](#)

Organizational structure and governance

Consider establishing an AI center of excellence and a model governance committee.

AI center of excellence

Establish an [AI center of excellence \(AI CoE\)](#) (AWS blog post) to guide generative AI initiatives across the organization. The AI CoE should offer guidance, best practices, and technical capabilities for building generative AI applications. Through regular engagement with business units, the AI CoE helps identify opportunities for generative AI adoption while maintaining governance and quality standards.

Model governance committee

It is essential to have a dedicated model governance committee that has clear roles and responsibilities. This committee develops evaluation criteria for foundation models, reviews usage requests, and validates compliance with ethical AI principles. Working closely with legal and compliance teams, the committee oversees model performance and risk assessments. This establishes a balanced approach to innovation and responsible AI use.

Standardization and technical excellence

Consider establishing a library of reusable generative AI patterns and tools, and also create an enterprise-wide access framework that democratizes access to AI resources.

Pattern library and tooling

Develop a suite of standardized tools and predefined patterns for common generative AI applications. Create a centralized repository that contains well-documented patterns, code templates, and architecture diagrams. This standardization lowers the barrier to entry, improves consistency across implementations, and accelerates development by providing clear starting points.

Enterprise-wide access framework

Facilitate organization-wide access to advanced tools, such as Amazon Q Business and Amazon Q Developer, through streamlined processes for requesting access, training, and onboarding. This democratization of AI resources empowers teams across departments while maintaining proper security controls and governance.

Process implementation

Implement processes that help you manage the following:

- Services
- Model selection and evaluation
- Performance monitoring and evaluation
- Security and compliance

Service management

Internal service management processes are crucial for organizing and controlling generative AI adoption. These processes should cover technical evaluation of new models, legal reviews, and access requests. By establishing clear workflows and responsibilities, organizations can maintain proper oversight while improving deployment and operations efficiency for generative AI solutions.

Model selection and evaluation

A systematic approach to model selection is critical for balancing capability, cost, and performance. Begin proof-of-concept development with top-tier models to quickly validate business value and gain stakeholder buy-in. After the use case is proven, systematically evaluate smaller models against established performance benchmarks to optimize costs for production. This approach accelerates initial development while promoting cost-effective scaling. This process requires clear communication about performance expectations and careful documentation of required capabilities.

Performance monitoring and optimization

Effective monitoring and optimization require a comprehensive approach to tracking both technical and business metrics. Develop dashboards to track key metrics, such as inference latency, throughput, error rates, and cost per inference. Set up alerts for anomalies or performance degradation. Conduct regular performance reviews to make sure that models continue to meet business needs.

Cost management should be proactive and strategic. Regular review of model usage patterns and compute resource optimization helps maintain efficient operations. Implement cost-allocation tags and budget monitoring to maintain visibility into expenses across different use cases and business units.

Security and compliance

Security and compliance considerations must be embedded throughout the generative AI implementation lifecycle. Develop a comprehensive risk management framework that addresses data privacy, model security, and ethical AI considerations. This framework should align with existing enterprise security policies and address the unique challenges of generative AI applications.

Implement security controls through a layered approach, starting with robust access management and authentication. Make sure that proper network security and data protection measures are in place, including comprehensive audit logging and monitoring capabilities. Establish clear incident response procedures that are specific to generative AI applications.

Implementation recommendations

For successful generative AI adoption across the enterprise, consider the following key recommendations:

- Start with well-defined, limited-scope projects that can demonstrate clear business value.
- Document success metrics and learnings to inform future projects.
- Scale successful implementations gradually, and make sure that proper controls and support mechanisms are in place.
- Maintain focus on continuous improvement through regular reviews of processes and procedures.
- Create comprehensive training programs that cover both technical and responsible AI practices.
- Establish clear mechanisms for knowledge sharing and cross-team collaboration.

Through careful attention to these best practices and recommendations, organizations can build a strong foundation for sustainable generative AI adoption while maintaining security, efficiency, and ethical considerations.

Next steps

Generative AI represents a transformative opportunity for enterprises to innovate, optimize operations, and create new value for customers. However, the journey from experimentation to enterprise-wide adoption requires a strategic approach that balances innovation with security, governance, and operational excellence.

The four-layer framework presented in this strategy—reliable infrastructure, approved foundation models, robust security and governance, and repeatable application patterns—provides organizations with a comprehensive approach to building and scaling their generative AI initiatives. This framework helps enterprises move beyond isolated prototypes to create sustainable, production-ready solutions that deliver measurable business value.

Success in enterprise generative AI adoption requires more than just technical implementation. Organizations must cultivate the right combination of people, processes, and technology. The establishment of an AI center of excellence, implementation of proper governance structures, and development of clear processes for model evaluation and deployment are as crucial as the technical infrastructure itself.

Security and governance cannot be afterthoughts in generative AI implementation. By embedding these considerations from the start, organizations can innovate while prioritizing the protection of sensitive data and compliance with regulatory requirements. The security controls and governance frameworks outlined in this strategy provide a foundation for responsible AI adoption that can evolve with changing requirements and emerging threats.

The application patterns presented in this strategy document demonstrate how organizations can accelerate their generative AI initiatives by using proven approaches. Whether implementing generative AI-powered assistants, RAG chat-based assistants, or intelligent document processing solutions, these patterns provide a starting point for rapid development while ensuring consistency and quality across implementations.

As organizations embark on their generative AI journey, consider taking the following next steps:

1. Assess your current AI and data infrastructure readiness against the framework presented in this strategy.
2. Establish cross-functional teams to oversee generative AI initiatives, including representatives from IT, data science, legal, and business units.

3. Start with small, well-defined proof-of-concept projects that can demonstrate clear business value.
4. Develop a comprehensive generative AI strategy that aligns with your organization's broader business objectives.
5. Invest in ongoing education and training to build generative AI capabilities across your organization.

Through the comprehensive suite of AWS services and with the solid foundation provided by Amazon Bedrock, organizations can confidently move from generative AI experimentation to scalable, enterprise-ready solutions that are designed to be secure, ethical, and effective. As you navigate this transformative journey, AWS remains committed to supporting your success with the tools, expertise, and infrastructure needed to unlock the full potential of generative AI in your enterprise.

Resources

The following resources provide additional context, technical details, and implementation guidance to support your organization's generative AI journey. As AWS continues to innovate with generative AI, you can regularly check these resources for updates and new capabilities.

AWS Prescriptive Guidance

Strategy resources

- [Generative AI workload assessment](#)
- [Maturity model for adopting generative AI on AWS](#)
- [Retrieval Augmented Generation options and architectures on AWS](#)
- [Prompt engineering best practices to avoid prompt injection attacks on modern LLMs](#)

RAG and chat-based assistants

- [Choosing an AWS vector database for RAG use cases](#)
- [Deploy a RAG use case on AWS by using Terraform and Amazon Bedrock](#)
- [Develop advanced generative AI chat-based assistants by using RAG and ReAct prompting](#)
- [Develop a fully automated chat-based assistant by using Amazon Bedrock agents and knowledge bases](#)

Coding assistants

- [Use Amazon Q Developer as a coding assistant to increase your productivity](#)

Document processing

- [Document institutional knowledge from voice inputs by using Amazon Bedrock and Amazon Transcribe](#)
- [Automatically extract content from PDF files using Amazon Textract](#)

AWS service documentation

- [Amazon Bedrock documentation](#)
- [Amazon SageMaker AI documentation](#)

Other AWS resources

- [Transform your business with generative AI](#) (AWS website)
- [Generative AI innovation center](#) (AWS website)
- [Securing generative AI: An introduction to the generative AI security scoping matrix](#) (AWS blog post)
- [Establishing an AI/ML center of excellence](#) (AWS blog post)
- [Architect defense-in-depth security for generative AI applications using the OWASP Top 10 for LLMs](#) (AWS blog post)
- [Setting up secure, well-governed machine learning environments on AWS](#) (AWS blog post)
- [Machine learning lens](#) (AWS Well-Architected Framework)

Other resources

- [OWASP top 10 for large language model applications](#) (OWASP website)

Document history

The following table describes significant changes to this guide.

Change	Description	Date
Initial publication	—	June 26, 2025